Johann Wolfgang von Goethe went on to have five children of his own, of whom all but the eldest son – August – died within two weeks of their birth. In all probability the cause was incompatibility between the blood groups of Goethe and his wife, Christiane, which after the first successful pregnancy led the mother to develop antibodies to the fetal blood. This condition, known as rhesus disease, is nowadays treated so effectively that the mortality rate is less than 2 per cent, but in the 1790s it had an average mortality rate of 50 per cent, and for Goethe's four younger children it was a death sentence.[8]

Altogether in Goethe's family – a well-to-do German family in the late eighteenth century – the child survival rate was an abysmal 25 per cent. Only three out of twelve children reached adulthood. This horrendous statistic was not exceptional. Around the time Goethe wrote 'The Sorcerer's Apprentice' in 1797, it is estimated that only about 50 per cent of German children reached age fifteen,[9] and the same was probably true in most other parts of the world.[10] By 2020, 95.6 per cent of children worldwide lived beyond their fifteenth birthday,[11] and in Germany that figure was 99.5 per cent.[12] This momentous achievement would not have been possible without collecting, analyzing and sharing massive amounts of medical data about things like blood groups. In this case, then, the naive view of information proved to be correct.

However, the naive view of information sees only part of the picture, and the history of the modern age was not just about reducing child mortality. In recent generations humanity has experienced the greatest increase ever in both the amount and the speed of our information production. Every smartphone contains more information than the ancient Library of Alexandria[13] and enables its owner to instantaneously connect to billions of other people throughout the world. Yet with all this information circulating at breathtaking speeds, humanity is closer than ever to annihilating itself.

Despite – or perhaps because of – our hoard of data, we are continuing to spew greenhouse gases into the atmosphere, pollute rivers and oceans, cut down forests, destroy entire habitats, drive countless species to extinction, and jeopardise the ecological foundations of our own species. We are also producing ever more powerful weapons of mass destruction, from thermonuclear bombs to doomsday viruses. Our leaders don't lack